

Production Order Create/Edit — Screen Processing Design (画面処理設計)

001_DD-SPD — elaborates 001_DD, implements 001_BD (SCR-001), requirements REQ-010–REQ-019.

Document control (改版履歴)

Field	Value
Document ID	001_DD-SPD
System name	ProductionManagementAI
Subsystem name	Production orders
Work item	WI-002
Created by	Claude (for ThanhTN)
Created date	2026-09-18
Last updated by	Claude (for ThanhTN)
Last updated date	2026-09-22

Version	Date	Author	Revision content
1	2026-09-18	Claude (for ThanhTN)	Initial creation. Moves 001_DD revision 1's processing flows P-01–P-05 into this document and splits them per component
2	2026-09-22	Claude (for ThanhTN)	WI-004 (DEC-016): the page renders inside the shared <code>AppHeader</code> , which now contains the navbar (<code>AppNavbar</code> , designed in 003_DD module 10 and 003_DD-SPD §8; breadcrumb links become <code>GuardedLink</code> , 003_DD module 11). No processing step of this screen changes. WI-004 DEC-022: the form registers its dirty state with the shared <code>NavigationGuard</code> (003_DD module 11, 003_DD-SPD §9), so navbar, breadcrumb and app-name links open the existing discard dialog with a pending destination (001_BD v7 E-07a); Discard navigates there
3	2026-09-23	Claude (for ThanhTN)	WI-005: the UI is Japanese. Page titles, button names and the dialog title quote the Japanese text with an English gloss (WI-005 DEC-003). No field, rule, API or processing changes

Overview and process list

Field	Value
Screen / file name	<code>src/frontend/src/features/production-orders/ProductionOrderPage.tsx</code> and the components it composes; the server side of each request is in <code>ProductionOrdersController</code>
Overview	How SCR-001 loads, validates, saves, cancels and reacts to every API outcome, one block per component

No	Process name	Overview	Notes
1	<code>ProductionOrderPage</code> — P-01 Load	Role gate, parallel fetch, page state, title, flash message	E-01
2	<code>ProductionOrderForm</code> — P-02 Create / P-03 Update (client side)	Blur/submit validation, request build, response handling	E-02–E-06, E-08
3	<code>ProductionOrderForm</code> — P-05 Save-response handling	Mapping of each HTTP outcome to UI	E-06
4	<code>DiscardChangesDialog</code> — P-04 Cancel	Dirty check and confirmation	E-07, E-09
5	<code>ProductionOrdersController</code> / <code>ProductsController</code> — request boundary	Auth, content type, dispatch to the service, result → HTTP	Server steps are in 001_DD-FN

Reference documents

No	Document	Purpose / use	Notes
1	001_DD	Items, states, message catalog, module list	Parent document
2	001_DD-API	Endpoint contracts called below	
3	001_DD-FN	Server-side steps (<code>CreateAsync</code> , <code>UpdateAsync</code> , ...)	
4	001_BD	Events E-01–E-09, validation V-01–V-08	

Processing design

1. ProductionOrderPage — P-01 Load

Field	Value
Detail	Route component for <code>/production-orders/new</code> and <code>/production-orders/:id</code>

Field	Value
Created by / date	Claude / 2026-09-18
Last modified by / date	—

Processing overview: decide the mode from the route, apply the client-side role gate, fetch the data, then either render the form or a full-page panel. Runs on mount and on **再読み込み** (Reload) / **Try again**.

Used components / services

No	Name	Overview	Notes
1	<code>useAuth</code>	Current user and roles	WI-001
2	<code>api.ts</code> (<code>listProducts</code> , <code>getOrder</code>)	Typed wrappers over <code>apiClient</code>	001_DD module 10
3	<code>ProductionOrderForm</code> , <code>MessageBanner</code>	Rendered when ready	
4	<code>useLocation</code> / <code>useNavigate</code>	Flash message from the create redirect	react-router

Processing flow

Step	Description	Branch / condition	Calls	Result (state / redirect / render)
1	Mode = <code>create</code> if the route is <code>/new</code> , else <code>edit</code> with <code>id</code>	—	—	—
2	Role gate on <code>user.roles</code>	Neither <code>Admin</code> nor <code>Operator</code> → stop	—	<code>forbidden</code> panel (MSG-E012), no API call
3	Set state <code>loading</code> , <code>document.title</code> to the mode's title (the order number is added after load)	—	—	Skeleton card, <code>aria-busy="true"</code>
4	Fetch in parallel: products, and in edit mode the order	—	<code>GET /api/products</code> ; <code>GET /api/production-orders/{id}</code>	—
5	Order response	404 → <code>notFound</code> ; 401 → <code>apiClient</code> redirects to <code>/login</code> ; 403 → <code>forbidden</code> ; network/5xx → <code>loadError</code>	<code>apiClient</code>	Not-found (MSG-E011) / forbidden / load-error (MSG-E013 + Try again) panel, focus moved to the panel
6	Products response	Failure → <code>loadError</code> ; empty list → continue with no options	<code>apiClient</code>	Product select shows MSG-E014 and is disabled

Step	Description	Branch / condition	Calls	Result (state / redirect / render)
7	Build <code>initialValues</code> : from the order (edit) or empty with status <code>Draft</code> (create); set title <code>製造指示 {orderNumber}</code> – <code>ProductionManagementAI</code> (Production order {orderNumber}) in edit mode; create mode's title is <code>新規製造指示 – ProductionManagementAI</code>	—	—	<code>ready</code>
8	If <code>location.state.flash</code> is set (after create), show that message and clear the state with <code>navigate(location.pathname, { replace: true, state: null })</code>	No flash → skip	—	Success banner MSG-I001

2. ProductionOrderForm — P-02 Create / P-03 Update (client side)

Field	Value
Detail	Controlled form for items 6–16, owning values, errors, dirty state and the save request
Created by / date	Claude / 2026-09-18
Last modified by / date	—

Processing overview: validate on blur for early feedback, validate everything on Save, then send one request. The server repeats every check (001_DD-FN), so the client checks are advisory. After a successful update the form re-bases on the server response, so `isDirty` becomes false and the lock/status options reflect the new status.

Used components / services

No	Name	Overview	Notes
1	<code>validation.ts</code>	<code>validateProduct</code> , <code>validateQuantity</code> , <code>validateDueDate(value, initial, mode)</code> , <code>validateNotes</code> → message ID or null	Mirrors V-01–V-05
2	<code>messages.ts</code>	Message ID → text	001_DD catalog
3	<code>api.ts</code> (<code>createOrder</code> , <code>updateOrder</code>)	POST/PUT wrappers	001_DD-API \$2, \$4
4	<code>FieldError</code> , <code>MessageBanner</code>	Error display	

Processing flow — blur (E-02) and status change (E-08)

Step	Description	Branch / condition	Calls	Result (state / redirect / render)
1	Field loses focus	Read-only field → skip	—	—

Step	Description	Branch / condition	Calls	Result (state / redirect / render)
2	Run that field's validator	Due date: check "≥ browser-local today" only in create mode or when the value differs from <code>initialValues.dueDate</code> (DEC-009, DEC-021)	<code>validation.ts</code>	Inline error with <code>aria-invalid</code> , or the error cleared
3	Notes input (E-03)	—	—	Counter <code>{n}/500</code> (code points); red above 500
4	Status change	Options are only <code>[current, ...allowedNextStatuses]</code>	—	Value changes locally; nothing saved until Save

Processing flow — Save (E-04)

Step	Description	Branch / condition	Calls	Result (state / redirect / render)
1	Run all validators (read-only fields skipped)	Any error → show all, focus the first invalid field in DOM order, stop	<code>validation.ts</code>	Validation-error state
2	Build the body: <code>productId</code> , <code>quantity</code> (as number), <code>dueDate</code> , <code>notes</code> (trimmed, "" → null); in edit mode also <code>status</code> and <code>version</code> from <code>initialValues</code>	—	—	—
3	Set <code>saving</code> : Save disabled, label 「保存中...」 (Saving...)	—	—	Saving state
4	Send	Create → POST; Edit → PUT	<code>createOrder</code> / <code>updateOrder</code>	—
5	Create success (201)	—	<code>navigate('/production-orders/{id}', { replace: true, state: { flash: 'MSG-I001' } })</code>	Page re-runs P-01 in edit mode; <code>replace</code> keeps Back from returning to an empty create form
6	Update success (200)	—	—	<code>initialValues = values = response</code> ; banner MSG-I002 (<code>role="status"</code>); lock and status options recomputed from <code>isProductQuantityEditable</code> / <code>allowedNextStatuses</code>
7	Failure	—	Process 3 (P-05)	—
8	Clear <code>saving</code>	always	—	Save enabled again

3. ProductionOrderForm — P-05 Save-response handling (E-06)

Field	Value
Detail	Maps every non-success save outcome to UI
Created by / date	Claude / 2026-09-18
Last modified by / date	—

Processing overview: an `ApiError` from `apiClient` carries the status and the parsed Problem Details. Entered values are always kept, except after a 401 (session gone) or when the user presses **Reload** after a 409.

Used components / services

No	Name	Overview	Notes
1	<code>apiClient</code> <code>ApiError</code>	<code>{ status, problem? }</code>	001_DD module 10
2	<code>messages.ts</code>	IDs → text	

Processing flow

Step	Description	Branch / condition	Calls	Result (state / redirect / render)
1	400 <code>VALIDATION</code>	For each key in <code>problem.errors</code> → that field's message	—	Inline errors, focus first invalid
2	401	—	<code>apiClient</code> <code>onUnauthorized</code>	Auth cleared → <code>ProtectedRoute</code> → <code>/login</code> (unsaved values lost; accepted)
3	403	—	—	Forbidden panel (MSG-E012)
4	404 <code>MSG-E011</code>	—	—	Not-found panel
5	409 <code>MSG-E009</code>	—	—	Error banner with 再読み込み (Reload) → P-01 (discards local edits)
6	422 <code>MSG-E007</code> / <code>MSG-E008</code>	—	—	Error banner with that message
7	415, 5xx, network error	—	—	Error banner MSG-E013; Save enabled again

4. DiscardChangesDialog — P-04 Cancel (E-07, E-09)

Field	Value
Detail	Confirmation before discarding unsaved changes (REQ-019, DEC-018)
Created by / date	Claude / 2026-09-18
Last modified by / date	—

Processing overview: Cancel checks `isDirty` (values compared with `initialValues`, notes trimmed). Only a dirty form opens the dialog. Covers the Cancel button only, not browser Back or tab close (DEC-018).

Used components / services

No	Name	Overview	Notes
1	Native <code><dialog></code> + <code>showModal()</code>	Focus trap, Escape, <code>::backdrop</code>	No library
2	<code>useNavigate</code>	Leave to <code>/production-orders</code>	SCR-002, since WI-003

Processing flow

Step	Description	Branch / condition	Calls	Result (state / redirect / render)
1	Cancel clicked	<code>isDirty</code> false → <code>navigate('/')</code>	—	Home
2	Open the dialog	<code>isDirty</code> true	<code>dialog.showModal()</code>	Dialog 「変更を破棄しますか？」 (Discard your changes?); focus on 編集を続ける (Keep editing)
3	破棄 (Discard)	—	<code>dialog.close()</code> , <code>navigate('/')</code>	Home; nothing saved
4	編集を続ける (Keep editing) or Escape (<code>cancel</code> event)	—	<code>dialog.close()</code>	Form with all values; focus returns to 「キャンセル」 (Cancel)

5. ProductionOrdersController / ProductsController — request boundary

Field	Value
Detail	HTTP entry points for 001_DD-API; no business logic

Field	Value
Created by / date	Claude / 2026-09-18
Last modified by / date	—

Processing overview: the ASP.NET Core pipeline authenticates and authorizes, then checks the content type. The action calls one service method and maps its `Result` to HTTP. All business checks happen in 001_DD-FN.

Used components / services

No	Name	Overview	Notes
1	Policy <code>ProductionOrderEditor</code>	<code>RequireRole("Admin", "Operator")</code>	DEC-001
2	<code>[Consumes("application/json")]</code>	415 on other content types	DEC-020
3	<code>ProductionOrderService</code>	Use cases	001_DD-FN
4	<code>ProblemResults</code> helper	<code>Result</code> → Problem Details (<code>type</code> , <code>title</code> , <code>status</code> , <code>code</code> , <code>errors</code> , <code>traceId</code>)	DEC-023

Processing flow

Step	Description	Branch / condition	Calls	Result (state / redirect / render)
1	Authentication (cookie)	None → 401, empty body	ASP.NET Core	—
2	Authorization (policy)	Missing role → 403, empty body	ASP.NET Core	—
3	Route match <code>{id:guid}</code>	Not a GUID → 404	routing	—
4	Content type (POST/PUT)	Not JSON → 415	<code>[Consumes]</code>	—
5	Model binding	Binding error → 400 with message IDs (custom <code>InvalidModelStateResponseFactory</code>)	MVC	—
6	Call the service method	—	<code>ListProductsAsync</code> / <code>GetAsync</code> / <code>CreateAsync</code> / <code>UpdateAsync</code>	—
7	Map the result	<code>Ok</code> → 200 (201 + <code>Location</code> for create); <code>Invalid</code> → 400; <code>NotFound</code> → 404; <code>Conflict</code> → 409; <code>RuleViolation</code> → 422	<code>ProblemResults</code>	Response

Step	Description	Branch / condition	Calls	Result (state / redirect / render)
8	Unhandled exception	—	<code>AddProblemDetails</code> + exception handler	500 MSG-E013, no exception detail

Unresolved decisions

None. Related decisions: DEC-007, DEC-008, DEC-009, DEC-018, DEC-020, DEC-021, DEC-023 in `work-items/WI-002/decisions.md`.